Identification of Key Mitochondrial Signatures in Alzheimer's Disease Across Human Postmortem and iPSC-Derived Neurons Models

Daiyun Dong[1], Waqar Ahmed[1,2], Ram Sagar[1,2], Vasiliki Machairaki[1,2]

1. Department of Genetic Medicine, Johns Hopkins University, School of Medicine, Baltimore, MD, USA

2. The Richman Family Precision Medicine Center of Excellence in Alzheimer's Disease, Johns Hopkins University School of Medicine, Baltimore, MD, USA

Introduction

Alzheimer's disease (AD) is a brain disorder that affects memory, thinking, and behavior[1]. Increasing evidence has implicated mitochondrial dysfunction in AD[2], and we aim to identify the key mitochondria-related AD biomarkers. We performed a comparative expression profiling analysis, incorporating seven AD human postmortem (HP) datasets from different brain regions and three iPSC-derived neurons (iNs) datasets. Through comparison, we identified nine mitochondria-related genes that exhibit consistent dysregulation in both human brains and iNs, suggesting key mitochondrial dysfunctions that contribute significantly to AD pathology.

Methods

We combined the p-values of genes from six HP datasets (Table 1) with Fisher's method[3], identifying HP common differentially expressed genes (DEGs) with an adjusted combined p-value < 0.05 and consistent dysregulated direction. Then we obtained the key DEGs between the HP and iN by overlapping the HP common DEGs and DEGs from three iN datasets (Table 1). MitoCarta 3.0[4] was used to annotate mitochondria-related DEGs (MitoDEGs) among these key DEGs. Finally, a random forest model was built using the seventh HP dataset (Table 1) to assess the predictive performance of key MitoDEGs for AD.

Workflow

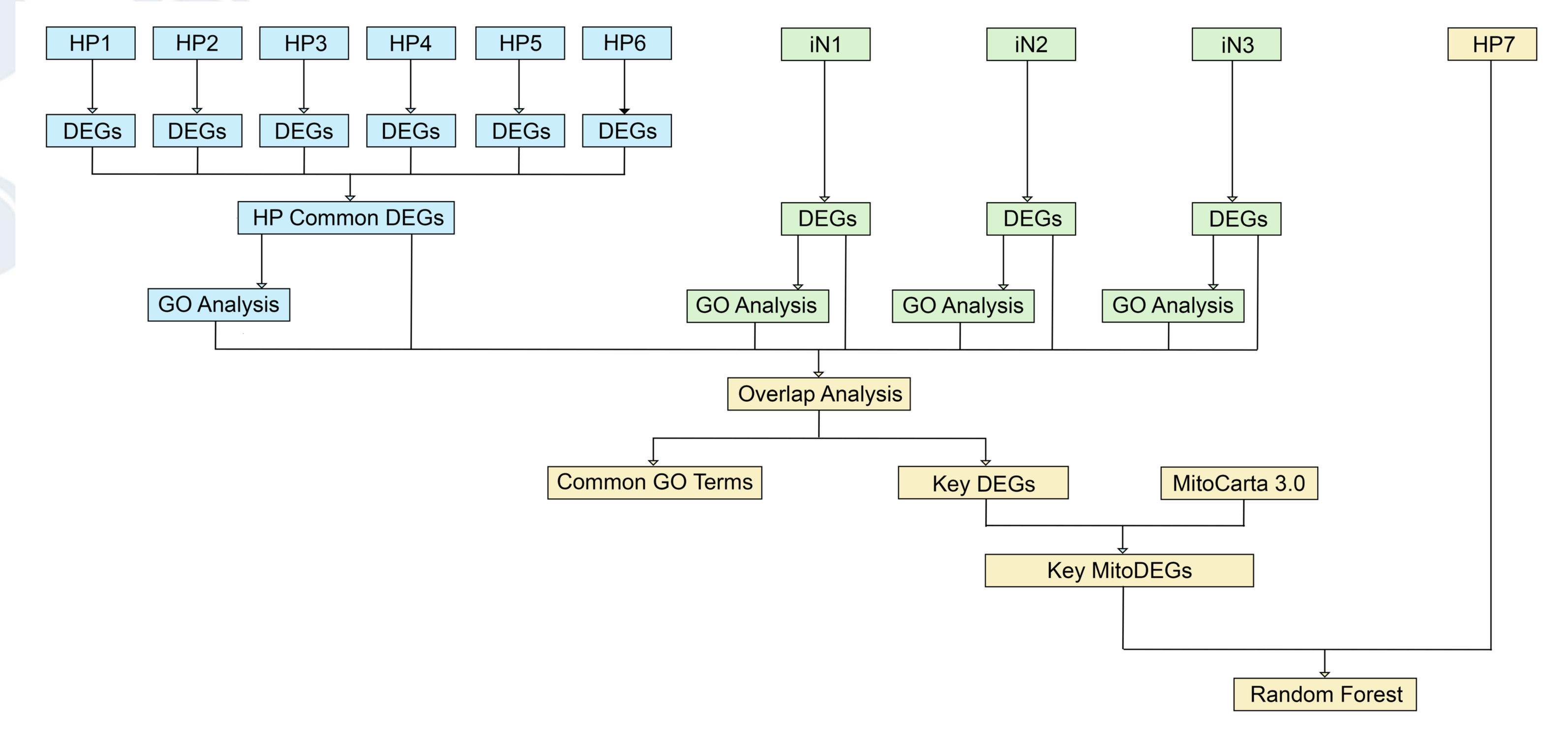

Figure 1. Workflow of Study Process.

Datasets

Table 1. Human Postmortem and iPSC-derived Neurons Datasets Used

Dataset	Source	Accession number
HP1	Human Postmortem - Middle Temporal Gyrus	GSE132903
HP2	Human Postmortem - Dorsolateral Prefrontal Cortex	GSE33000
HP3	Human Postmortem - Visual Cortex	GSE44771
HP4	Human Postmortem - Entorhinal Cortex	GSE118553
HP5	Human Postmortem - Temporal Cortex	GSE118553
HP6	Human Postmortem - Middle Temporal Cortex	GSE5281
HP7	Human postmortem - Prefrontal Cortex	GSE44770
iN1	iPSC Derived Neurons - Familial AD (APP V717I)	GSE231639
iN2	iPSC Derived Neurons - Familial AD (PSEN1 A246E)	GSE95673
iN3	iPSC Derived Neurons - Sporadic AD	GSE162873

These datasets can be accessed in NCBI Gene Expression Omnibus (GEO) database.

Results

We identified a total of 9,837 HP common DEGs (5,433 upregulated, 4,404 downregulated), among which 472 were mitochondria-related genes (167 upregulated, 305 downregulated)(Fig 2). We then intersected this HP common DEGs list with the DEGs from the three iN datasets (Fig 3A), identified a total of 173 common DEGs (142 upregulated, 31 downregulated), including 9 consistently upregulated mitochondria-related DEGs (Fig 3B).

Mitochondria-related HP Common DEGs

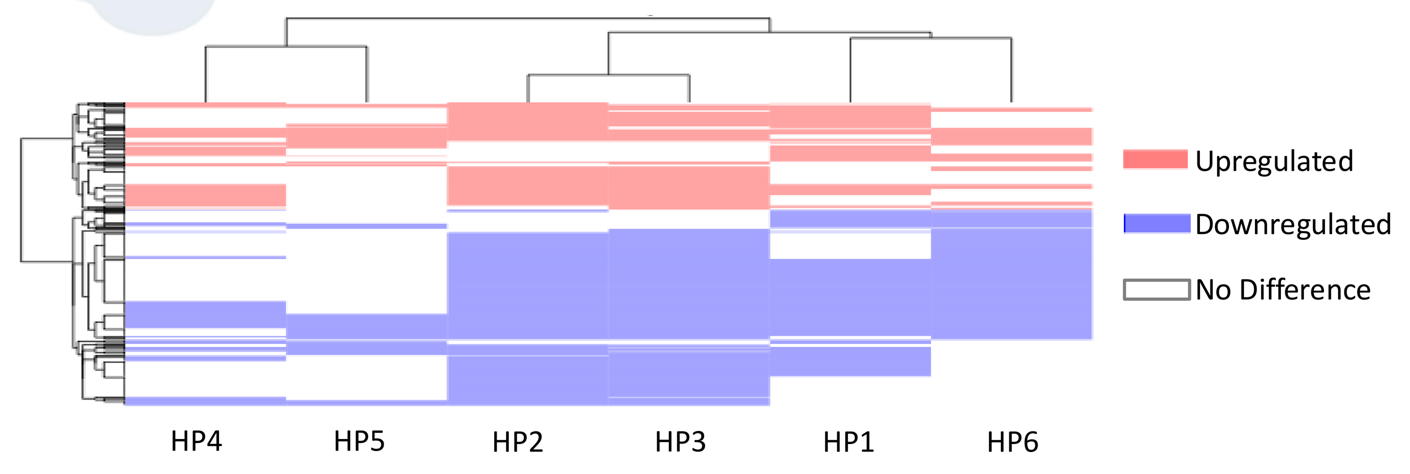

Figure 2. Expression of Mitochondria-related HP Common DEGs in Six HP Datasets. Red indicates significant upregulation in AD, blue indicates significant downregulation in AD, and white indicates no significant differences. (Threshold: Padj < 0.05)

Nine Common MitoDEGs

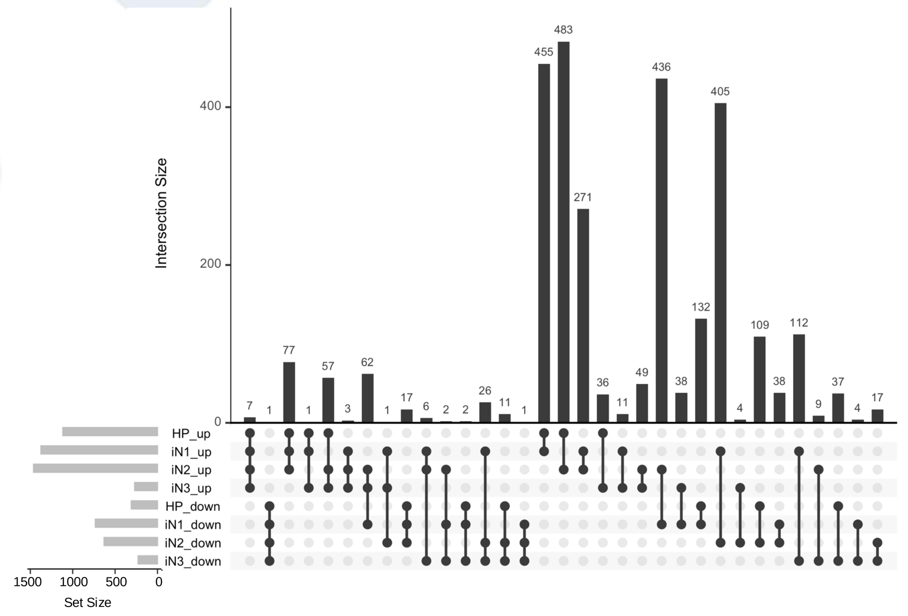

Figure3A. Overlap of HP common DEGs and iN DEGs. DEGs were split according to their dysregulated direction. We selected mitochondria-related genes that were differentially expressed in the HP and at least two iN DEG lists, with a consistent dysregulated direction.

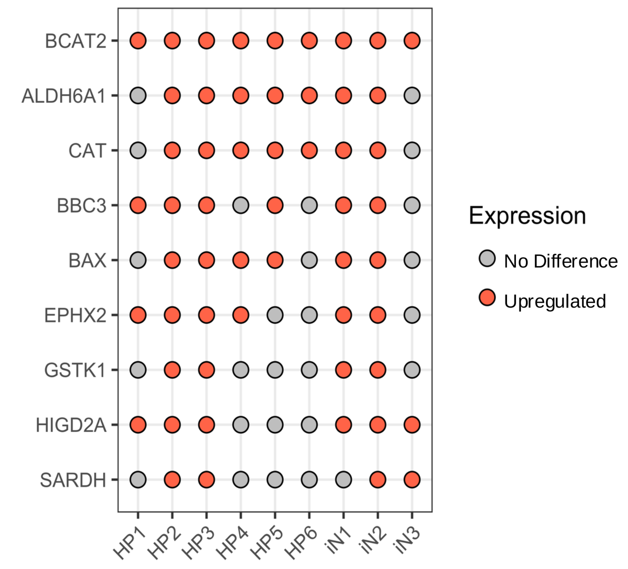

Figure 3B. Expression of MitoDEGs in HP and iN Datasets. Red indicates significant upregulation in AD, blue indicates significant downregulation in AD, and white indicates no significant differences. (Threshold: Padj < 0.05)

We constructed a random forest model using the nine common MitoDEGs and seventh HP dataset, selecting 1,000 as the number of decision trees. The mean AUC of the 5-fold cross-validation is 0.91 (95% CI: 0.88 - 0.94) (Fig 4A), indicating that the model based on these nine common mitoDEGs perform well in distinguishing AD patients from the control group. Among nine genes, BCAT2 showed the highest importance in the model (Fig 4B). Function analysis indicates that these nine genes are highly related to redox reactions and Reactive Oxygen Species (ROS) accumulation (Fig 5A), while eight of them are located in the mitochondria (Fig 5B).

Machine Learning Validation of Nine MitoDEGs

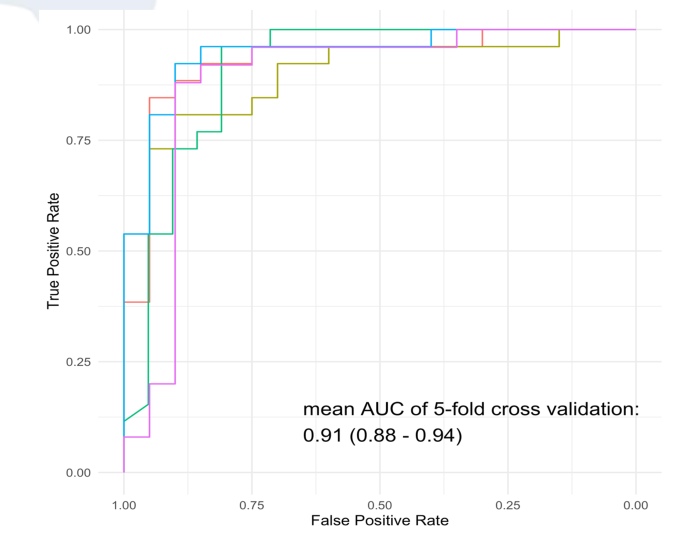

Figure 4A. ROC Curve for The Classification Model between AD and Controls. The mean AUC of the 5-fold cross-validation is 0.91 (95% CI: 0.88 - 0.94).

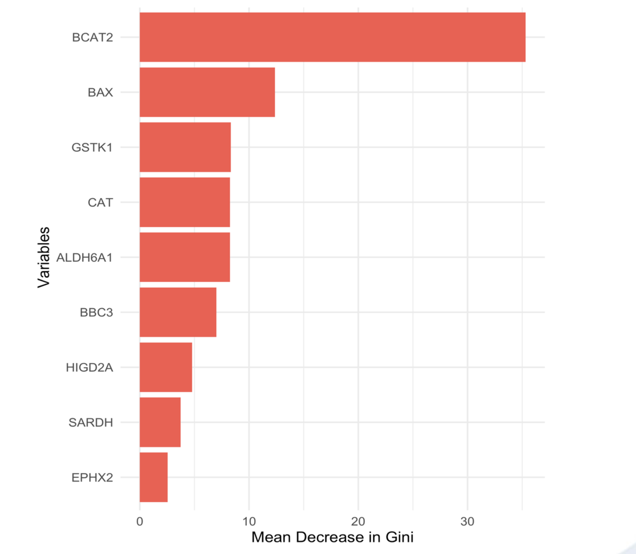

Figure 4B. The Importance of Variables Measured by the Mean Decrease Gini Coefficients. BCAT2 shows the most important among nine MitoDEGs.

Location and Function of Nine MitDEGs

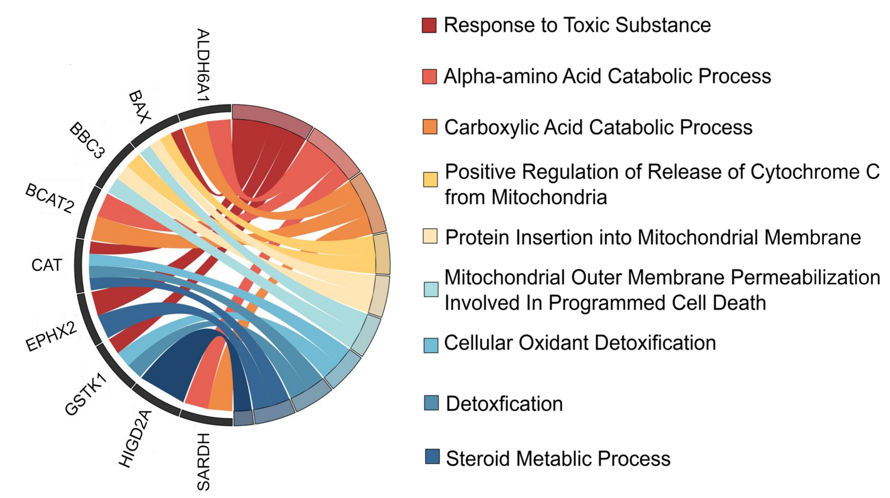

Figure 5A. The Biological Process of Nine MitoDEGs. The nine MitoDEGs are related to oxidation-reduction reactions and apoptosis.

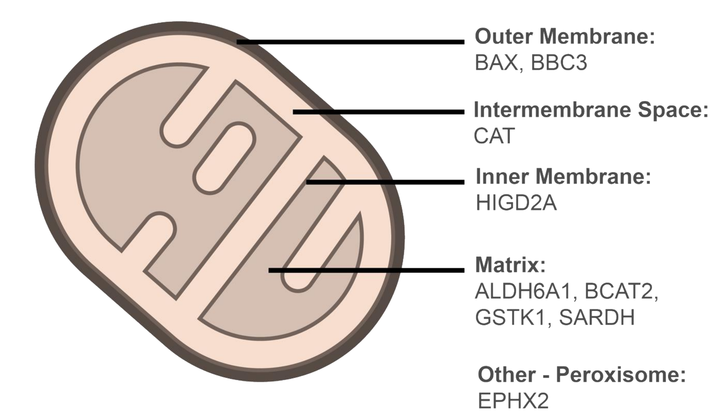

Figure 5B. The Cellular Location of Nine MitoDEGs. Eight of them are located in the mitochondria, including mitochondrial outer membrane, mitochondrial intermembrane space, mitochondrial inner membrane, and mitochondrial matrix.

Conclusions

- In the brain tissues of AD patients and iPSC-derived neurons derived from them, there is widespread dysregulation of mitochondria-related genes.
- Nine genes are potential markers for AD prognosis and treatment.
  - GSTK1, CAT, ALDH6A1, and EPHX2 directly regulate redox balance.
  - BAX, BBC3, and HIGD2A promote apoptosis, leading to ROS accumulation.
  - BCAT2 is involved in branched-chain amino acid metabolism, and SARDH is involved in one-carbon metabolism; their metabolic products and reactions are associated with redox pathways.

References:

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